The leximin core selection on exact games

Draft fragment: boundary theorems, exact bump paths, and the open crux

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Status: mathematical core of a paper in progress. Singleton bumps are proved monotone even on balanced games. Grand-coalition bumps remain on an exact path but labeled resource monotonicity is disproved by an exact five-player example. Proper nonsingleton bumps are reduced to an exact capacity dichotomy and one open path lemma; the capacity implication is proved for every two-level failure dual, including all two-player coalitions, and a sharp quantitative carrier–star inequality is proved for two-player coalitions. Balanced nonexact counterexamples establish that exactness is necessary for the remaining positive program.

1 The leximin-above-floors core selection

Let (N, v) be a balanced TU game, $M = v(N)$, and floors $m_i := v(\{i\})$. For $x \in \mathbb{R}^N$ write the *gains* $g_i(x) = x_i - m_i$. The *leximin-above-floors* selection is

$$\phi(v) := \arg \operatorname*{leximin}_{x \in \mathrm{Core}(v)} \big(g_1(x), \dots, g_n(x)\big),$$

the unique core allocation whose ascending-sorted gain vector is lexicographically largest. It is single-valued (leximin over a polytope) and is computed by the Maschler-type scheme over player coordinates: repeatedly raise a common lower level s of the free players' gains as far as the core allows, fix ("pin") every free player whose gain cannot exceed $m_i + s$ on the level-optimal face, and recurse on the rest. On *exact* games $m_i = \min_{\mathrm{Core}} x_i$, so $g_i \geq 0$ and ϕ egalitarianly distributes the surplus $x_i - v(\{i\})$ inside the core; on convex games it is the constrained egalitarian solution of Dutta and Ray (1989). Implementation (exact rational + float) in `src/leximin.py`, validated against those two coincidences.

We ask whether ϕ is *coalitionally monotonic*: if $v' = v + d\,\mathbf{1}_S$ for a single coalition S and $d > 0$, is $\phi_i(v') \geq \phi_i(v)$ for every $i \in S$? A positive answer on exact games would exhibit an explicit coalitionally monotonic core selection on that class. The reduction in Section 5 addresses only proper nonsingleton coalitions. Singleton bumps change the floors defining ϕ, and a grand-coalition bump changes the efficiency hyperplane. The singleton case is resolved next. The grand-coalition case has a negative answer even on a simple exact five-player game (Theorem 3.2).

2 Singleton bumps are monotone

We first isolate an elementary property of the leximin order. Write $a \succ_{\mathrm{lx}} b$ when the ascending rearrangement of a is lexicographically larger than that of b.

Lemma 2.1 (one-coordinate translation)**.** *If $a \succ_{\mathrm{lx}} b$, $a_i < b_i$, and $d > 0$, then $a + de_i \succ_{\mathrm{lx}} b + de_i$.*

Proof. For a finite vector z, put $L_\lambda(z) = \sum_j e^{-\lambda z_j}$. The first unequal coordinate of the ascending rearrangements dominates this sum as $\lambda \to \infty$; hence $a \succ_{\mathrm{lx}} b$ if and only if $L_\lambda(a) < L_\lambda(b)$ for all sufficiently large λ. Moreover,

$$L_\lambda(b + de_i) - L_\lambda(a + de_i) = L_\lambda(b) - L_\lambda(a) + (1 - e^{-\lambda d})(e^{-\lambda a_i} - e^{-\lambda b_i}).$$

Both terms on the right are positive for all sufficiently large λ, because $a_i < b_i$. The same characterization gives the claim. □

Theorem 2.2 (singleton coalitional monotonicity)**.** *Let v and $v' = v + d\,\mathbf{1}_{\{i\}}$ be balanced games, with $d > 0$. Then*

$$\phi_i(v') \geq \phi_i(v).$$

Thus singleton bumps are monotone even without exactness.

Proof. Let $m_j = v(\{j\})$ and express the old core in gain coordinates

$$G = \{(x_j - m_j)_{j \in N} : x \in \mathrm{Core}(v)\}.$$

If the singleton floor of player i is raised by d, the new feasible gain set is

$$H = \{g - de_i : g \in G,\ g_i \geq d\}.$$

Let z be the leximin point of G and h the leximin point of H. The desired inequality is $h_i + d \geq z_i$. If $z_i < d$, this is immediate from $h_i \geq 0$. Suppose $z_i \geq d$. Then $b = z - de_i$ belongs to H, so h is either b or $h \succ_{\mathrm{lx}} b$. If, contrary to the claim, $h_i + d < z_i$, then $h_i < b_i$ and the equality case is impossible. Lemma 2.1 gives

$$h + de_i \succ_{\mathrm{lx}} b + de_i = z.$$

But $h + de_i \in G$, contradicting the definition of z as the leximin point of G. □

3 Grand-coalition bumps: an exact path but no resource monotonicity

The grand-coalition case also has an exact path, but its geometry differs from the nested-core path of Section 5.

Lemma 3.1 (grand-bump exact path)**.** *Let v be exact and, for $r \geq 0$, let v^r agree with v on every proper coalition and satisfy $v^r(N) = v(N) + r$. Then v^r is exact for every $r \geq 0$. Moreover,*

$$\mathrm{Core}(v) + r\Delta_N \subseteq \mathrm{Core}(v^r), \qquad \Delta_N = \{a \in \mathbb{R}^N_+ : a(N) = 1\}.$$

Proof. If $x \in \mathrm{Core}(v)$ and $a \in r\Delta_N$, then $x + a$ has total $v(N) + r$ and weakly raises every coalition sum, proving the inclusion and balancedness. Fix a proper coalition T. Exactness of v supplies $x^T \in \mathrm{Core}(v)$ with $x^T(T) = v(T)$. Choose $j \notin T$. Then $x^T + re_j \in \mathrm{Core}(v^r)$ by the inclusion and remains tight at T. Thus

$$\min_{x \in \mathrm{Core}(v^r)} x(T) = v(T) = v^r(T).$$

The grand coalition is tight throughout its core by efficiency, so v^r is exact at every coalition. □

The exact path does not rescue labeled resource monotonicity.

Theorem 3.2 (grand resource monotonicity fails on exact games)**.** *There is an exact five-player game v such that, when only $v(N)$ is raised, one player's leximin-above-floors payoff strictly falls. Indeed, let*

$$p = (0, 0, 7, 0, 7), \qquad q = (2, 10, 0, 2, 0),$$

put $v(N) = 14$, and for every proper T put $v(T) = \min\{p(T), q(T)\}$. If $v^r(N) = 14 + r$ and $v^r(T) = v(T)$ for $T \subsetneq N$, then for $0 \le \varepsilon \le 3/2$,

$$\phi(v^{2+\varepsilon}) = \left(3 + \frac{\varepsilon}{3}, 4 - \frac{\varepsilon}{3}, 3 + \frac{\varepsilon}{3}, 3 + \frac{\varepsilon}{3}, 3 + \frac{\varepsilon}{3}\right).$$

Thus player 2 falls at rate $1/3$ while the grand worth rises. For example, raising the grand worth from 16 to $129/8$ changes that player's payoff from 4 to $95/24$.

Proof. Both p and q have total 14 and belong to $\mathrm{Core}(v)$. At every proper coalition one of them attains $v(T)$, so v is exact; every v^r is exact by Lemma 3.1. All singleton worths are zero.

Write $s = 3+\varepsilon/3$ and $z = (s, 7-s, s, s, s)$. Direct substitution in the lower-envelope definition of v shows that $z \in \mathrm{Core}(v^{2+\varepsilon})$ throughout the stated interval. (At $\varepsilon = 0$ every core slack is nonnegative; the only coalition sum with negative slope is the singleton of player 2, whose slack remains positive. Every other slope is nonnegative.)

It remains to identify the leximin point. The two pair constraints

$$x_2 + x_3 \ge 7, \qquad x_2 + x_5 \ge 7$$

belong to the core. If every coordinate of a core point is at least t, then efficiency and the first constraint give $16 + \varepsilon = x(N) \ge 7 + 3t$, hence $t \le s$. So z attains the largest possible minimum coordinate. On that level-optimal face put $d_i = x_i - s \ge 0$ and $D = 1 - 2\varepsilon/3 \ge 0$. Efficiency gives $\sum_i d_i = D$, while the two displayed constraints give $d_2 + d_3 \ge D$ and $d_2 + d_5 \ge D$. Nonnegativity forces $d_2 = D$ and every other $d_i = 0$. Hence the level-optimal face is the singleton $\{z\}$, proving the formula and the strict drop. □

The counterexample pinpoints the failure mechanism: four players form the lowest leximin stratum, and additional total resource raises them jointly only by taking payoff from a higher-stratum player pinned to two cross-stratum pair constraints. The feasible comparison points supplied by $\mathrm{Core}(v) + r\Delta_N$ therefore do not control labeled coordinates.

4 Exactness is necessary

Coalitional monotonicity of ϕ fails on balanced games in general—already at $n = 4$, and even at $n = 5$ where every floor is core-attained. So exactness cannot be weakened to balancedness.

Theorem 4.1. *There is a balanced (non-exact) five-player game v, a coalition S with $|S| = 3$, and $d > 0$ with $v + d\,\mathbf{1}_S$ balanced, such that some $i \in S$ has $\phi_i(v + d\,\mathbf{1}_S) < \phi_i(v)$. The same holds for a four-player game.*

Example 4.2 ($n = 5$; floors attained)**.** *Let $N = \{1, \dots, 5\}$ with $v(N) = 7$ and (all other coalitions as listed in the supplement) in particular*

$$v(\{1,2,3\}) = 3, \quad v(\{2,3\}) = 3, \quad v(\{1,3,5\}) = v(\{3,5\}) = 4, \quad v(\{1,5\}) = -1.$$

Floors are $m = (0,1,1,1,2)$*, and* $m_i = \min_{\mathrm{Core}} x_i$ *for* every i *(all floors attained); the game is balanced but not exact (e.g.* $\min_{\mathrm{Core}} x(\{1,5\}) = 1 > -1$*). Then*

$$\phi(v) = \left(\tfrac{1}{2}, \tfrac{3}{2}, \tfrac{3}{2}, 1, \tfrac{5}{2}\right), \qquad \phi(v + \mathbf{1}_{\{1,2,3\}}) = (1,1,2,1,2),$$

both in their respective cores, and player $2 \in \{1,2,3\}$ drops *from* $\frac{3}{2}$ *to* 1 *when* $v(\{1,2,3\})$ *is raised from* 3 *to* 4*. Thus exactness at the singletons (floor attainment) does not suffice: it is exactness at higher coalitions that the positive theorem needs.*

Example 4.3 ($n = 4$; floor unattained)**.** *Let* $N = \{1,2,3,4\}$*,* $v(N) = 20$*, with* $v(\{1\}) = -3$*,* $v(\{1,2\}) = 11$*,* $v(\{1,3\}) = 7$*,* $v(\{2,3\}) = 4$*,* $v(\{1,2,3\}) = 13$*,* $v(\{2,4\}) = 10$*,* $v(\{1,3,4\}) = 9$*,* $v(\{2,3,4\}) = 13$*, and the remaining values in the supplement. Here* $v(\{1\}) = -3 < 1 = \min_{\mathrm{Core}} x_1$*. Then* $\phi(v) = (\frac{5}{2}, \frac{17}{2}, \frac{9}{2}, \frac{9}{2})$ *and* $\phi(v + 3\,\mathbf{1}_{\{1,2,3\}}) = (2,9,5,4)$*: player* $1 \in \{1,2,3\}$ *drops from* $\frac{5}{2}$ *to* 2*.*

Both examples are certified in exact rational arithmetic (ϕ in the core on both sides, the drop exact); see the supplement and `src/hunt_leximin_nonexact.py`.

5 Reduction for proper nonsingleton bumps on exact games

Fix an exact v, a proper coalition S with $|S| \geq 2$, and $d > 0$ with $v' = v + d\,\mathbf{1}_S$ exact. Since $|S| \geq 2$ the floors are unchanged, and

$$\mathrm{Core}(v') = \mathrm{Core}(v) \cap \{x : x(S) \geq v(S) + d\} \subseteq \mathrm{Core}(v).$$

For $\tau \in [v(S), v(S)+d]$ let v_τ agree with v except $v_\tau(S) = \tau$, so $\mathrm{Core}(v_\tau) = \mathrm{Core}(v) \cap \{x(S) \geq \tau\}$ interpolates $\mathrm{Core}(v)$ and $\mathrm{Core}(v')$.

Lemma 5.1 (exact interval)**.** *Every* v_τ*,* $\tau \in [v(S), v(S)+d]$*, is exact.*

Proof. $\mathrm{Core}(v_\tau) \supseteq \mathrm{Core}(v') \neq \emptyset$, so v_τ is balanced. Exactness needs $\min_{\mathrm{Core}(v_\tau)} x(T) = v_\tau(T)$ for all T. For $T = S$, exactness of v supplies $x^0 \in \mathrm{Core}(v)$ with $x^0(S) = v(S)$, while any $x^1 \in \mathrm{Core}(v')$ satisfies $x^1(S) \geq v(S) + d$. The core of v is convex, so the segment $[x^0, x^1]$ contains a point $x^\tau \in \mathrm{Core}(v)$ with $x^\tau(S) = \tau$. This point belongs to $\mathrm{Core}(v_\tau)$, proving $\min_{\mathrm{Core}(v_\tau)} x(S) = \tau$. For $T \neq S$: put $f_T(\tau) = \min_{\mathrm{Core}(v_\tau)} x(T)$. As τ grows $\mathrm{Core}(v_\tau)$ shrinks, so f_T is nondecreasing; and $\mathrm{Core}(v_\tau) \subseteq \mathrm{Core}(v)$ gives $f_T(\tau) \geq \min_{\mathrm{Core}(v)} x(T) = v(T)$. At the endpoints $f_T = v(T)$ because v and v' are exact. A nondecreasing function equal to $v(T)$ at both endpoints and $\geq v(T)$ between is $\equiv v(T)$. Hence $f_T(\tau) = v(T) = v_\tau(T)$. □

The entire exact-bump range can be read from exposed core faces. For $T \subseteq N$, write $F_T = \{x \in \mathrm{Core}(v) : x(T) = v(T)\}$ (with $F_N = \mathrm{Core}(v)$), and define

$$d_v^*(S) = \min_{T \neq S} \left(\max_{x \in F_T} x(S) - v(S) \right).$$

Proposition 5.2 (exact bump capacity)**.** *For an exact game* v *and a proper coalition* S*, the bumped game* $v + d\mathbf{1}_S$ *is exact if and only if* $0 \leq d \leq d_v^*(S)$*. If* $\sigma_S = \phi(v)(S) - v(S)$*, then every exact bump at* S *is non-binding if and only if* $d_v^*(S) \leq \sigma_S$*; a binding exact bump exists if and only if* $d_v^*(S) > \sigma_S$*.*

Proof. The bumped core is $K_d = \mathrm{Core}(v) \cap \{x : x(S) \geq v(S) + d\}$. For $T \neq S$, exactness at T is equivalent to $F_T \cap K_d \neq \emptyset$, which is equivalent to

$$d \leq \max_{x \in F_T} x(S) - v(S).$$

Taking the minimum over $T \neq S$ gives all the higher-coalition exactness conditions; the term $T = N$ also gives nonemptiness of K_d. Exactness at S follows by interpolating between a point of F_S and a point of K_d, as in Lemma 5.1. This proves the first statement. The last two statements compare the exact capacity with the largest bump still satisfied by the old selection, namely σ_S. □

The capacity implication can be proved for every star-failure dual with at most two residual levels. This includes every two-player coalition and also explains the same-stratum pair certificates found in the computation.

Proposition 5.3 (two-level dual certificate)**.** *Let v be exact, S proper, $\phi = \phi(v)$, and $\sigma_S = \phi(S) - v(S)$. Suppose $(\star_S)$ fails and its dual maximal face can be represented by weights taking at most two values on S: after normalization, for some $\emptyset \neq C \subsetneq S$ and $R > 1$,*

$$\phi \in \arg \max_{x \in \mathrm{Core}(v)} \left(x(C) + R x(S \setminus C)\right).$$

Then

$$\max_{x \in F_C} x(S) \leq \phi(S) \quad \textit{and hence} \quad d_v^*(S) \leq \sigma_S.$$

The same conclusion holds in the one-level case where ϕ maximizes $x(S)$.

Proof. Put $H = S \setminus C$ and $h = \phi(C) - v(C) \geq 0$. Exactness makes $F_C = \{x \in \mathrm{Core}(v) : x(C) = v(C)\}$ nonempty. For $x \in F_C$, maximality of ϕ gives

$$(x(C) - \phi(C)) + R(x(H) - \phi(H)) \leq 0.$$

The first parenthesis is $-h$, so $x(H) - \phi(H) \leq h/R$. Therefore

$$x(S) - \phi(S) = -h + x(H) - \phi(H) \leq -h\left(1 - \frac{1}{R}\right) \leq 0.$$

Thus F_C certifies the required bound:

$$d_v^*(S) \leq \phi(S) - v(S) = \sigma_S.$$

In the one-level case, $\phi(S) = \max_{\mathrm{Core}(v)} x(S)$, so the conclusion is immediate. □

Corollary 5.4 (the capacity implication for pairs)**.** *If v is exact and S is a proper two-player coalition, then*

$$d_v^*(S) > \sigma_S \quad \Longrightarrow \quad (\star_S).$$

Proof. When $(\star_S)$ fails, its positive dual weights on the two members of S have either one value or two. Normalize the smaller value to one and apply Proposition 5.3. □

There is a second exact certificate, independent of the number of residual levels. It compares the slack of a coalition carrying the whole complement of S with the slack of its trace inside S.

Proposition 5.5 (off-side slack-domination certificate)**.** *Let v be exact, S proper, $\phi = \phi(v)$, and $P = N \setminus S$. Suppose C is a nonempty proper subset of S, put $T = P \cup C$, and assume*

$$\phi(T) - v(T) \le \phi(C) - v(C).$$

Then

$$\max_{x \in F_C} x(S) \le \phi(S), \qquad \text{so} \qquad d_v^*(S) \le \phi(S) - v(S).$$

In particular, the conclusion holds whenever T is ϕ-tight.

Proof. Because $S \cup T = N$ and $S \cap T = C$, modularity gives $x(S) + x(T) = x(N) + x(C)$ for every allocation x. If $x \in F_C$, then core feasibility at T and efficiency imply

$$x(S) = v(N) + v(C) - x(T) \le v(N) + v(C) - v(T).$$

Moreover,

$$v(N) + v(C) - v(T) = \phi(S) + \big(\phi(T) - v(T)\big) - \big(\phi(C) - v(C)\big) \le \phi(S).$$

Exactness makes F_C nonempty, so this face occurs in the minimum defining $d_v^*(S)$ and proves the claim. □

The carrier comparison has a useful normal form that removes both slack expressions. It also converts failure of every candidate into a single finite-dimensional separation problem.

Proposition 5.6 (carrier marginal and face separation)**.** *Let v be exact, $\phi = \phi(v)$, S proper, and $P = N \setminus S$. For every $k \in S$,*

$$\begin{aligned}[\phi(P \cup \{k\}) - v(P \cup \{k\})] - [\phi_k - v(\{k\})] \\ = \phi(P) - [v(P \cup \{k\}) - v(\{k\})].\end{aligned}$$

Consequently k satisfies the carrier inequality if and only if

$$v(P \cup \{k\}) - v(\{k\}) \ge \phi(P). \tag{M}$$

Let $K \subseteq S$ be nonempty. If (M) fails strictly for every $k \in K$, then every point of the exact face $F_{P \cup \{k\}}$ satisfies $x(S) > \phi(S)$. If, in addition, $(\star_S)$ fails, there is a nonzero $w \in \mathbb{R}_+^S$ such that

$$\max_{k \in K} \max_{x \in F_{P \cup \{k\}}} \sum_{i \in S} w_i (x_i - \phi_i) < 0. \tag{Sep}$$

Proof. Because P and $\{k\}$ are disjoint, $\phi(P \cup \{k\}) = \phi(P) + \phi_k$, which proves the identity and (M). Now suppose the latter fails for k and take $x \in F_{P \cup \{k\}}$. Core feasibility at $\{k\}$ gives

$$x(P) = v(P \cup \{k\}) - x_k \le v(P \cup \{k\}) - v(\{k\}) < \phi(P).$$

Efficiency therefore gives $x(S) > \phi(S)$.

Let Z be the convex hull of the union of the faces $F_{P \cup \{k\}}$, $k \in K$. It is a compact subset of the core, and every point of Z has S-sum strictly larger than $\phi(S)$. Hence failure of $(\star_S)$ implies that Z is disjoint from $\phi_S + \mathbb{R}_+^S$ after projection onto the S-coordinates. Strong separation gives a strict separating normal w. Its coordinates must be nonnegative, since otherwise its infimum on $\phi_S + \mathbb{R}_+^S$ would be $-\infty$. Translating by ϕ_S gives (Sep). □

The separator is in fact forced by the star-failure cover; it is not an independent normal that can be eliminated against that cover.

Proposition 5.7 (residual-gap separator)**.** *Suppose failure of* $(\star_S)$ *has a tight-coalition certificate*

$$r^S = \mu \mathbf{1}_N - \sum_T \lambda_T \mathbf{1}_T, \qquad \lambda_T \geq 0, \quad \phi(T) = v(T) \text{ if } \lambda_T > 0,$$

where $r_i \geq 1$ *on* S*. Put*

$$r_0 = \min_{i \in S} r_i, \qquad K = \{i \in S : r_i = r_0\}, \qquad H = S \setminus K, \qquad D = \sum_{i \in H} (r_i - r_0).$$

Assume $H \neq \emptyset$*. If the carrier marginal fails at* $k \in K$ *by*

$$\gamma_k = [\phi(P \cup \{k\}) - v(P \cup \{k\})] - [\phi_k - v(\{k\})] > 0,$$

then, for every $x \in F_{P \cup \{k\}}$*,*

$$\sum_{i \in H} \frac{r_i - r_0}{D} (x_i - \phi_i) \leq -\frac{r_0}{D} \gamma_k < 0. \tag{RG}$$

Consequently the normalized residual gap is a common strict separator for every failed candidate in K*, supported entirely on the higher-residual class* H*.*

Proof. For any $x \in \mathrm{Core}(v)$ put $d = x - \phi$. Efficiency gives $d(N) = 0$, and core feasibility plus tightness gives $d(T) \geq 0$ whenever $\lambda_T > 0$. Taking the scalar product of the cover identity with d therefore yields

$$\sum_{i \in S} r_i d_i = -\sum_T \lambda_T d(T) \leq 0.$$

Splitting off the minimum residual gives

$$r_0 d(S) + \sum_{i \in H} (r_i - r_0) d_i \leq 0.$$

For $x \in F_{P \cup \{k\}}$, the carrier calculation in Proposition 5.6 gives $d(S) \geq \gamma_k > 0$. Substitution and division by D prove (RG). □

Corollary 5.8 (residual-gap escape criterion)**.** *Under the hypotheses of Proposition 5.7, if some* $k \in K$ *satisfies*

$$\max_{x \in F_{P \cup \{k\}}} \sum_{i \in H} \frac{r_i - r_0}{D} (x_i - \phi_i) \geq 0, \tag{E}$$

then k *satisfies the carrier inequality*

$$[\phi(P \cup \{k\}) - v(P \cup \{k\})] \leq [\phi_k - v(\{k\})].$$

Thus existence of a minimum-residual carrier is reduced to escape from one explicit scalar separator; no coordinatewise choice of a member of K *is required.*

Proof. The contrapositive is exactly inequality (RG). □

The strict separator also has a finite dual form. This removes the quantification over all points of the carrier-tight faces and exposes the precise slack identity induced by the residual cover.

Proposition 5.9 (dual form of the carrier-face separator)**.** *Use the hypotheses and notation of Proposition 5.6, and put $T_k = P \cup \{k\}$. A vector $w \in \mathbb{R}^S_+$ satisfies (Sep) if and only if, for every $k \in K$, there are multipliers $\theta^k_U \geq 0$ on the nonempty proper coalitions and $a_k, b_k \in \mathbb{R}$ such that*

$$w^S = a_k \mathbf{1}_N + b_k \mathbf{1}_{T_k} - \sum_{\emptyset \neq U \subsetneq N} \theta^k_U \mathbf{1}_U, \tag{D1}$$

where w^S is w on S and zero on P, and

$$\sum_U \theta^k_U \left[\phi(U) - v(U)\right] < b_k[\phi(T_k) - v(T_k)]. \tag{D2}$$

Whenever the carrier marginal fails at k, the right-hand slack in (D2) is positive and every such certificate has $b_k > 0$.

Proof. Fix k and maximize $w^S \cdot x$ over the core subject to $x(T_k) = v(T_k)$. In the convention $-x(U) \leq -v(U)$ for the core inequalities, the linear-programming dual has nonnegative multipliers θ^k_U, free multipliers a_k, b_k on efficiency and the face equality, and stationarity exactly (D1). Its objective is

$$-\sum_U \theta^k_U v(U) + a_k v(N) + b_k v(T_k).$$

Taking the scalar product of (D1) with ϕ shows that this objective is strictly below $w \cdot \phi_S$ exactly when (D2) holds. Strong duality therefore proves the equivalence, simultaneously for all k because the same w occurs in each stationarity identity.

If the carrier marginal fails, then $\phi(T_k) - v(T_k) > \phi_k - v(\{k\}) \geq 0$. The left side of (D2) is nonnegative, so strictness forces $b_k > 0$. □

Corollary 5.10 (aggregated separator identity)**.** *If the carrier marginal fails for every $k \in K$, then the separator in Proposition 5.6 supplies $c > 0$, $A \in \mathbb{R}$, a collection $\eta_U \geq 0$, and a nonzero $w \in \mathbb{R}^S_+$ such that*

$$\mathbf{1}_K = c w^S - A \mathbf{1}_N - |K| \mathbf{1}_P + \sum_U \eta_U \mathbf{1}_U, \tag{AD1}$$

and

$$\sum_U \eta_U [\phi(U) - v(U)] < \sum_{k \in K} [\phi(P \cup \{k\}) - v(P \cup \{k\})]. \tag{AD2}$$

More precisely, one may take

$$c = \sum_{k \in K} \frac{1}{b_k}, \qquad A = \sum_{k \in K} \frac{a_k}{b_k}, \qquad \eta_U = \sum_{k \in K} \frac{\theta^k_U}{b_k}.$$

Proof. Proposition 5.9 gives $b_k > 0$ for every $k \in K$. Divide (D1) and (D2) by b_k and sum over k. Since $\mathbf{1}_{P \cup \{k\}} = \mathbf{1}_P + e_k$ and $\sum_{k \in K} e_k = \mathbf{1}_K$, the stationarity sum rearranges to (AD1); the strict slack inequalities sum to (AD2). □

The same comparison has a quantitative converse at the level of total S-payoff. For a core point ψ, define

$$\gamma_S(\psi) = \min_{\emptyset \neq C \subsetneq S} \{[\psi(P \cup C) - v(P \cup C)] - [\psi(C) - v(C)]\},$$
$$\eta_S(\psi) = \max\{x(S) - \psi(S) : x \in \mathrm{Core}(v),\ x_i \geq \psi_i\ (i \in S)\}.$$

Thus $(\star_S)$ at ψ is equivalent to $\eta_S(\psi) > 0$.

Lemma 5.11 (exact carrier lift)**.** *If v is exact, then for every nonempty proper $C \subset S$ there is a $z^C \in \mathrm{Core}(v)$ such that*

$$z^C(S) - \psi(S) \geq [\psi(P \cup C) - v(P \cup C)] - [\psi(C) - v(C)].$$

Proof. Choose z^C tight at $P \cup C$. Core feasibility at C gives $z^C(P) \leq v(P \cup C) - v(C)$. Efficiency and the identity $\psi(P) = v(N) - \psi(S)$ now give the displayed inequality. □

The only possible obstruction is coordinatewise: the carrier-tight points raise the total payoff of S, but different members of S may fall at different points. For a pair, convexity removes that obstruction without loss.

Proposition 5.12 (sharp quantitative pair bound)**.** *If v is exact, $\psi \in \mathrm{Core}(v)$, and S is a proper two-player coalition, then*

$$\eta_S(\psi) \geq \gamma_S(\psi).$$

The coefficient one is sharp.

Proof. The claim is immediate if $\gamma_S(\psi) \leq 0$. Write $S = \{i, j\}$ and suppose $\gamma_S(\psi) > 0$. Exactness supplies core points a, b tight at $N \setminus \{j\}$ and $N \setminus \{i\}$, respectively. Tightness makes $a_j = \max_{x \in \mathrm{Core}(v)} x_j \geq \psi_j$ and $b_i = \max_{x \in \mathrm{Core}(v)} x_i \geq \psi_i$, while Lemma 5.11 gives

$$a(S) - \psi(S) \geq \gamma_S(\psi), \qquad b(S) - \psi(S) \geq \gamma_S(\psi).$$

If either point dominates ψ on both coordinates, it proves the claim. Otherwise $a_i < \psi_i$ and $b_j < \psi_j$. The segment $[a, b]$ meets the northeast orthant based at (ψ_i, ψ_j): writing the two displacements as (u_i, u_j) and (w_i, w_j), positive total gives $u_j > -u_i > 0$ and $w_i > -w_j > 0$, hence $u_j w_i > (-u_i)(-w_j)$, exactly the condition that the two intervals of mixing weights making each coordinate nonnegative overlap. Every point on the segment has S-displacement at least $\gamma_S(\psi)$. The intersection point is therefore feasible in the definition of $\eta_S(\psi)$ and proves the bound. Exact lower-envelope examples attain equality. □

The slack certificate is automatic on the most important thin-core model class used by the searches.

Proposition 5.13 (two-generator lower envelopes)**.** *Let $p, q \in \mathbb{R}^N$ have the same total, and define the exact game*

$$v(T) = \min\{p(T), q(T)\} \quad (T \subsetneq N), \qquad v(N) = p(N) = q(N).$$

Let S be proper. If $\max_{x \in \mathrm{Core}(v)} x(S) > \phi(S)$ and $(\star_S)$ fails at ϕ, then there is a nonempty proper $C \subset S$ satisfying the hypothesis of Proposition 5.5. Consequently $d_v^*(S) \leq \phi(S) - v(S)$.

Proof. Both p and q belong to $\mathrm{Core}(v)$, and one of them attains $v(T)$ at every proper T, so v is exact. Put $P = N \setminus S$. Exactness at P gives

$$\max_{x \in \mathrm{Core}(v)} x(S) = v(N) - v(P) = \max\{p(S), q(S)\}.$$

After interchanging p and q, binding therefore gives $p(S) > \phi(S)$. The two generator sums on S cannot tie, since then $v(S) = \max_{x \in \mathrm{Core}(v)} x(S)$; hence $p(S) > q(S)$ and $p(P) < q(P)$. Define

$$C = \{i \in S : p_i > \phi_i\}.$$

This set is nonempty because $p(S) > \phi(S)$, and it is proper because otherwise $p \in \mathrm{Core}(v)$ would itself witness $(\star_S)$. For $i \in C$, core feasibility at the singleton gives

$$p_i > \phi_i \geq v(\{i\}) = \min\{p_i, q_i\},$$

so $q_i \leq \phi_i < p_i$. Hence

$$p(C) - q(C) \geq p(C) - \phi(C) \geq p(S) - \phi(S) = \phi(P) - p(P).$$

In particular $q(C) < p(C)$, and therefore

$$\begin{aligned} v(P \cup C) - v(C) &= \min\{p(P) + p(C), q(P) + q(C)\} - q(C) \\ &= \min\{p(P) + p(C) - q(C), q(P)\} \\ &\geq \phi(P). \end{aligned}$$

The last inequality also uses $\phi(P) \leq v(N) - v(S) = q(P)$, since $q(S) < p(S)$. Equivalently, $\phi(P \cup C) - v(P \cup C) \leq \phi(C) - v(C)$, which is precisely the off-side slack-domination hypothesis. □

So the bump is realized by a continuous path of *exact* games in which only the lower bound τ on $x(S)$ rises. This isolates the role of exactness: if endpoint exactness is weakened to balancedness, the analogous interior path need not be exact, and that is precisely the room inhabited by the counterexamples of Section 4. For proper nonsingleton bumps on exact games, coalitional monotonicity is therefore equivalent to the one-parameter statement:

Lemma 5.14 (monotone path — OPEN)**.** *Along the exact path v_τ, the map $\tau \mapsto \phi_i(v_\tau)$ is nondecreasing for every $i \in S$ (and, empirically, nonincreasing for $i \notin S$).*

Two reductions cut the work down further. First, the common case is trivial:

Lemma 5.15 (non-binding)**.** *If $\phi(v)(S) \geq v(S) + d$ then $\phi(v') = \phi(v)$.*

Proof. $\phi(v)$ is the leximin point of $\mathrm{Core}(v) \supseteq \mathrm{Core}(v')$ and lies in $\mathrm{Core}(v')$, hence is the leximin point of the smaller set too; ϕ is single-valued. □

Binding ($\phi(v)(S) < v(S) + d$) requires $\phi(v)(S) < \max_{\mathrm{Core}} x(S)$ and is empirically rare. In every binding case observed (thousands of exact-preserving bumps at $n \leq 6$, zero violations) one finds $\phi(v')(S) = v(S) + d$ and the two-sided pattern $\phi_i(v') \geq \phi_i(v)$ for $i \in S$, $\leq$ for $i \notin S$.

Table 1: Computational evidence (float LP, exact-certifiable) for the reduction. $(\star_S)$ evaluated at ϕ of the base game; "binding" bumps only.

statement	sample	violations
ϕ monotone for sampled proper exact bumps (mixed sizes)	2000+ bumps, $n \le 6$	0
(D) exact $\Rightarrow (\star_S)$ at ϕ	39/39, 13/13	0
monotone $\Rightarrow (\star_S)$	all balanced	0
$(\star_S)$ along the path $\Rightarrow$ monotone	600	0
$(\star_S)$ at one endpoint $\Rightarrow$ monotone	1696 balanced	22 non-exact

A cone criterion. For $x = \phi(w)$ with tight set $\mathcal{T}(x) = \{T : x(T) = w(T)\}$, say $(\star_S)$ holds if there is $e \in \mathbb{R}^N$ with

$$e(N) = 0, \quad e(S) > 0, \quad e_i \ge 0 \ \ (i \in S), \quad e(T) \ge 0 \ \ (T \in \mathcal{T}(x)) :$$

a first-order direction raising $x(S)$ without lowering any member of S or breaking a tight core constraint. Empirically $(\star_S)$ is *necessary* for the bump to be monotone (it fails at $\phi(v)$ in Example 4.2, which is why player 2 must drop), and exactness supplies it. This suggests the skeleton

(A) Lemma 5.1: the bump path is exact;

(D) each exact v_τ satisfies $(\star_S)$ at $\phi(v_\tau)$;

(E′) $(\star_S)$ along the whole path implies Lemma 5.14,

whose composition is the theorem. The supporting evidence is summarized in Table 1; (D) and (E′) remain open, and $(\star_S)$ alone is *not* sufficient (the last row), so a proof of (E′) must use exactness along the whole path, not merely $(\star_S)$ at one endpoint.

The open problem, sharply. A Farkas dual turns (D) into a statement about a single maximal face of the core. For a core point y written as a mixture of core points, $y(T) \ge v(T) = \phi(T)$ at every tight T, so $e = y - \phi$ meets the tight-cone part of $(\star_S)$ automatically; hence

$$\begin{aligned}(\star_S) &\iff \exists\, y \in \mathrm{Core}(v) : \ y_i \ge \phi_i \ (i \in S) \text{ and } y(S) > \phi(S) \\ &\iff \max\{y(S) : y \in \mathrm{Core}(v),\, y_i \ge \phi_i \ (i \in S)\} > \phi(S).\end{aligned}$$

Dualizing (verified in 200/200 random balanced cases, exact-certifiable):

$(\star_S)$ *fails* at $\phi(v)$ if and only if $\phi(v)$ maximizes

$$\sum_{k \in S} r_k \, x_k$$

over $\mathrm{Core}(v)$ for some weights $r_k \ge 1$ on S and zero weights outside S.

Equivalently, the tight coalitions $\{T : \phi(T) = v(T)\}$ carry nonnegative weights and a $\beta \ge 1$ covering each $k \notin S$ exactly β and each $k \in S$ at most $\beta - 1$. Thus **Lemma 5.14's hypothesis (D) is exactly the assertion that an exact game's leximin point never lies on a**

positive-S-weighted maximal face of its core at a binding-bumpable S. Uniform weights $r_k \equiv c$ are already excluded, since maximizing $c\,x(S)$ would force $\phi(S) = \max_{\mathrm{Core}} x(S)$, contrary to binding-bumpability; *the entire open content is that exactness excludes the non-uniform such faces.* The non-exact counterexamples of Section 4 realize exactly such faces—this is why $(\star_S)$ fails there and a member of S drops. (A corner scan for a deficient S-member pinned at its core-maximum found no occurrence in any exact binding case, $0/16^{+}$.)

The natural certificate is provably wrong. Where $(\star_S)$ *does* fail, one would like to prove $d_v^*(S) = 0$ and thereby certify directly that S is not exactly bumpable. This target is false. From the failure certificate, take the cover $\{\lambda_U\}$ dual to the failure, $W = \bigcup\{U : \lambda_U > 0\}$, and propose $C = W \cap S$. This route is closed by a six-player exact game (supplement, `WITNESS_KEY_FALSE_n6`): $(\star_S)$ fails at $S = \{1,3,4,5\}$ with cover $\{2,3\},\{1,2,6\},\{3,4,6\}$, yet a core point x with $x(W \cap S) = v(W \cap S)$ attains $x(S) = 24 > \phi(S) = 71/3$, so $W \cap S$ certifies nothing; the associated slack inequality $\phi(W) - v(W) \le \phi(W \cap S) - v(W \cap S)$ fails by $1/3$. Such failures are rare ($\approx 0.1\%$ of $(\star_S)$-failures at $n \in \{6,7\}$, found only under generators biased toward thin cores with nested tight chains), which is why sampling first suggested the cover certificate was universal. In every observed failure, however, the operative bound survives: there is a coalition C with $M^{(C)} := \max\{x(S) : x \in \mathrm{Core}(v),\, x(C) = v(C)\} \le \phi(S)$, so that S cannot be bumped past $\phi(S)$ while preserving exactness (in a $\approx 0.2\%$ minority $d_v^*(S) > 0$ — exact six-player witnesses with $d^* = 1$ and $d^* = 2$ are included in the supplement — but always $d_v^*(S) \le \phi(S) - v(S)$). Empirically the certificate has sharp structure: C can always be taken as a subset of S contained in a *single stratum* of the leximin computation — almost always a singleton. If C is ϕ-tight, then $\phi \in F_C$, so a certificate can only be boundary-sharp, $M^{(C)} = \phi(S)$; exact boundary cases of this kind do occur, while every strict certificate is necessarily non-tight. The open problem is thus sharpened: produce, from the structure of a $(\star_S)$-failure at the leximin point of an exact game, a single-stratum $C \subseteq S$ with $M^{(C)} \le \phi(S)$. Empirically the certificate is pinned further: writing r for the residuals of the failure's dual cover ($r_i \ge 1$ on S), a *minimum-residual* member $k \in S$ (one with $r_k = 1$) certifies in 99.6% ($n = 6$) to 100% ($n = 7$) of observed failures, the residue requiring a same-stratum pair. By strong duality the singleton form is equivalent to exhibiting nonnegative weights on ϕ-tight coalitions with coverage β on $N \setminus S$, $\beta - 1$ on $S \setminus \{k\}$, and at least $\beta - 1$ on k; weights supported inside S suffice in only about two thirds of cases, so a proof must construct this dual cover from ϕ-tight coalitions *crossing* S. The residual normal gives a possible stronger route, but not an equivalent reformulation. For a minimum-residual k, normalized by $r_k = 1$, and any core x with $x_k = v(\{k\})$, maximality of ϕ gives

$$[x(S) - \phi(S)] + \sum_{i \in S \setminus \{k\}} (r_i - 1)(x_i - \phi_i) \le 0.$$

Consequently the singleton face certificate would follow from the sufficient inequality

$$\sum_{i \in S \setminus \{k\}} (r_i - 1)(x_i - \phi_i) \ge 0$$

throughout that face. The sign is essential; the opposite inequality gives no upper bound on $x(S) - \phi(S)$. Whether this stronger face inequality is forced by the leximin strata is a separate question from the carrier route. An exact audit answers that stronger question negatively. A highest-gain minimum-residual player satisfies the nonnegative-correction condition in 139/141 saved multilevel failures, but not in two seven-player cases. In one of them

(WITNESS_KEY_CX_n7_s303_1491) the correction is identically $-3/2$ on each of the two highest-gain singleton faces, although both faces certify and both carrier inequalities hold. Thus the residual normal alone cannot prove the multilevel branch; a proof must use additional exactness geometry such as the off-side carrier comparison.

A new proof-producing exact audit isolates a still sharper target on the saved hard frontier. It covers all 141 stored three-or-more-level failures (106 at $n = 6$ and 35 at $n = 7$). In every case some minimum-residual singleton certifies, and, more strongly, some minimum-residual k satisfies the elementary carrier inequality

$$[\phi(P \cup \{k\}) - v(P \cup \{k\})] \le [\phi(\{k\}) - v(\{k\})].$$

Across the 367 individual minimum-residual candidates, 282 singleton faces certify and the carrier inequality explains 280 of them; it never produces a false certificate. The remaining two seven-player singleton certificates require a richer face dual, but another minimum-residual player in the same case has the carrier certificate. This is exact finite evidence, not a theorem. Nor can it be promoted to an unconditional singleton-carrier conjecture: the exact two-level witness above has no certifying singleton; its minimum-residual pair is essential. At this stage the apparent frontier was a dichotomy. A failure dual with at most two residual levels is already handled by Proposition 5.3. On the genuinely three-or-more-level branch, these data first suggested proving the displayed singleton inequality. The whole-class audit below supersedes that target with a canonical coalition K and removes the singleton-selection problem. The saved corpus initially suggested a canonical gain selector: in all 141 cases a carrier-certifying k could be chosen to maximize $\phi_i - v(\{i\})$ inside the minimum-residual class. That refinement is false. A seeded mutation audit produced 534 further exactly certified multilevel records (505 distinct game–coalition pairs). Some minimum-residual carrier still exists in all 534 records, but a highest-gain carrier exists in only 520, and a highest-gain singleton face certifies in only 533.

Here is the sharp selector counterexample. Let the exact six-player game be the lower envelope of the four common-sum points

$$p = (-6, 4, 18, -4, 11, -10), \quad q = (-1, 4, 24, -4, 0, -10),$$
$$a = (-1, -2, 19, -4, 6, -5), \quad b = (-6, 0, 20, 0, 9, -10),$$

and take $S = \{1, 2, 3, 4, 6\}$. Its leximin point is $\phi = (-3, 0, 20, -2, 6, -8)$, the normalized failure cover has three residual levels and minimum-residual class $K = \{1, 2, 3\}$, and the singleton gains on K are $(3, 2, 2)$. Thus player 1 is the unique highest-gain candidate. Nevertheless its carrier slack is $4 > 3$, and more strongly

$$\max\{x(S) : x \in \mathrm{Core}(v),\ x_1 = v(\{1\})\} = 8 > 7 = \phi(S),$$

so even its singleton face fails. Players 2 and 3 both satisfy the carrier inequality with equality and certify. Tight-hypergraph centrality and cover-weighted adjacency refinements also fail on exact mutations. The Any singleton route must therefore remain existential: select *some* minimum-residual player, without imposing a gain ranking. After adding fresh three-, four-, and five-generator screens and deduplicating all corpora, this existential carrier statement holds for all 669 distinct exact multilevel game–coalition pairs (702 certified records). The whole-class carrier statement below supersedes the need to solve this singleton-selection problem.

Proposition 5.6 sharpens what remains to be proved. With $K = \{k \in S : r_k = 1\}$, a counterexample would produce simultaneously the positive-S failure normal r and carrier failure at

every member of K. Proposition 5.7 shows that the resulting common separator is not a second obstruction to eliminate: it is automatically the normalized residual gap on $S \setminus K$. Thus mere coexistence of the two normals cannot yield a contradiction. A tempting coordinatewise strengthening is also false: exactness and the leximin strata need not permit a convex combination of the carrier-tight faces whose average displacement is nonnegative at every higher-residual coordinate. In the exact six-player instance `WITNESS_KEY_CX_n6_s201_10919`, the best common coordinatewise margin over all such mixtures is $-1/20$, although players 4 and 5 both satisfy the carrier inequality (each with margin $-3/4$).

The scalar condition in Corollary 5.8 is the minimal sufficient statement for a *fixed* failure cover, and it initially survives all 141 saved hard multilevel failures. Across their 367 minimum-residual faces, 277 escape; all 87 failed carriers are strictly negative, while three valid carriers do not escape but another candidate does in the same record.

Adversarial mutation shows that even this fixed-cover target is not universal. Among 505 distinct exact mutation cases, all have a minimum-residual carrier but only 123 have an escape face for the cover returned by the exact failure dual. Already trial 0 of the first seeded face-selector mutation audit has $K = \{1, 2, 3\}$ and normalized residual-gap weights $(2/3, 1/3)$ on players $5, 6$. Players 1 and 2 satisfy the carrier inequality, with margins -2 and 0, yet their face maxima are both $-1/3$; player 3 fails the carrier inequality and has maximum $-4/3$. Consequently neither coordinatewise neutralization nor escape for an arbitrarily chosen failure cover can be the missing theorem. A residual-based proof would have to select a favorable cover from the full dual face jointly with the carrier candidate; otherwise it must return to a cover-independent exactness exchange.

The cover-independent audit reveals a substantially simpler target. It uses the full minimum-residual class at once.

Conjecture 5.16 (minimum-residual-class carrier)**.** *Let v be exact and let $(\star_S)$ fail at ϕ. If the failure is not already covered by Proposition 5.3, then there is a failure cover with minimum-residual class $K = \{i \in S : r_i = r_0\}$ and $P = N \setminus S$ such that*

$$[\phi(P \cup K) - v(P \cup K)] \leq [\phi(K) - v(K)], \tag{KC}$$

or equivalently

$$v(P \cup K) - v(K) \geq \phi(P).$$

This single inequality would close the multilevel capacity branch immediately: apply Proposition 5.5 with $C = K$. It requires no singleton selector and is insensitive to how payoffs are ranked inside K. A deduplicated exact audit verifies (KC) for all 669 stored multilevel game–coalition pairs (702 records): 644 inequalities are strict and 25 are equalities, with margins ranging from -28 to 0. This includes all 505 adversarial mutation pairs on which the fixed-cover escape test mostly fails. Thus (KC), rather than a singleton rule or residual-gap escape, is now the sharp computational theorem target. A separate targeted screen of 3,000 four-generator lower-envelope games found 1,123 star failures and 25 genuinely multilevel failures, with no positive class-carrier margin; the closest boundary case was recomputed exactly and has margin zero. The broader screen is discovery evidence only, whereas the 669-pair audit is fully exact.

Proposition 5.3 explains an important part of this structure exactly. If the residuals have only the levels 1 and $R > 1$, the entire minimum-residual class $C = \{i \in S : r_i = 1\}$ is itself the required face certificate; no crossing-cover construction is needed. The rare same-stratum pair exceptions to singleton certification are examples with $|C| = 2$. Accordingly, all unresolved

certificate geometry lies in duals with at least three distinct residual levels. Proposition 5.5 removes a second large class: it is enough that an off-side carrier have no more slack than its trace inside S; tightness of the carrier is a particularly simple sufficient condition. In particular, the three-level witness `WITNESS_KEY_FALSE_n6` is certified by $T = \{1, 2, 6\}$ and $C = \{1\}$ (player labels one-indexed), even though the union-of-cover certificate described above fails.

Proposition 5.2 removes the frequency language from the logical frontier. Every pair (v, S) falls into the exact dichotomy

$$d_v^*(S) \le \sigma_S \quad \text{or} \quad d_v^*(S) > \sigma_S.$$

The first side admits no binding exact bump and is finished by Lemma 5.15. On the second side a binding exact bump exists, and the missing structural statement is precisely

$$d_v^*(S) > \sigma_S \quad \Longrightarrow \quad (\star_S).$$

Proposition 5.3 proves this implication whenever the failure dual has at most two residual levels, and Corollary 5.4 covers every $|S| = 2$. The remaining capacity frontier therefore requires both $|S| \ge 3$ and a genuinely three-or-more-level dual. After Proposition 5.5, a counterexample to the capacity implication must additionally have no off-side slack-dominating carrier. The singleton/pair certificate statistics motivate how to attack those cases, but neither their observed percentages nor sampling rarity enters the conditional theorem.

The sharpened computational target is correspondingly narrow. The broad slack-domination condition covered all 141 saved hard star-failure cases at $n = 6, 7$ and all 744 fresh binding star failures in a screen of exact lower-envelope games at $n = 4, 5, 6$, with no exception. The new class-carrier audit identifies the canonical coalition in the genuinely multilevel branch: the full minimum-residual class K works in all 669 distinct exact pairs. This remains evidence, not a theorem, but proving Conjecture 5.16 would close that branch with a single application of Proposition 5.5. Proposition 5.13 proves the broader existence of some slack-dominating carrier for every two-generator lower-envelope game. Thus within the finite lower-envelope model class, any counterexample must use at least three genuinely active generators; this is a structural restriction, not a sampling observation.

The obvious three-generator escape from the sharp two-generator boundary is itself blocked, but for an instructive reason. In the exact equality fixture, let $S = \{1, 2, 3, 4, 6\}$ and transfer $a = 1/10$ from the off-side player 5 of the second generator equally among the five members of S, retaining both original generators. All three generators are genuinely active. The leximin point remains $(1, 4, 3, 5, 7, 4)$ and the minimum off-side slack comparison becomes $1/50 > 0$, so the carrier condition has indeed been crossed. However, $(\star_S)$ is simultaneously restored: the core point

$$(26/25, 4, 153/50, 253/50, 171/25, 4)$$

dominates ϕ on S and has S-sum $429/25 > 17$. Thus the star opening is $4/25$, eight times the carrier escape. This is an exact boundary calculation, not a general theorem; a finite simplex-grid deformation screen found the same coupling throughout the tested local family. It suggests proving the carrier conjecture contrapositively: show that escaping every off-side carrier inequality constructs a core direction nonnegative on S, rather than trying to track one distinguished minimizing generator.

The carrier lift suggests a sharp quantitative form of that contrapositive:

$$\eta_S(\phi) \ge \gamma_S(\phi). \tag{Q}$$

We leave (Q) as a conjecture for exact games. Proposition 5.12 proves it for $|S| = 2$, and the coefficient one is already sharp there. In six exact lower-envelope screens covering 3,650 games and 35,629 binding pairs (v, S), all 746 cases with $\gamma_S(\phi) > 0$ satisfied (Q); the minimum in every screen was recomputed exactly and equalled one. These counts are evidence only. The exactness hypothesis is load-bearing even for the coefficient: a balanced non-exact five-player game has leximin point $(-3/2, 10, 6, 17/2, -10)$ and, at $S = \{2, 3, 4\}$, $\gamma_S = 7/2$ but $\eta_S = 1/2$. Thus the theorem target is a convex-combination statement for the carrier-tight points supplied by Lemma 5.11, not an inequality valid on arbitrary balanced cores.

Remark 5.17. *Lemma 5.14 is a leximin parametric-monotonicity statement; its cousin the nucleolus is* not *coalitionally monotone, so exactness (Lemma 5.1) is doing real work and the "self-defeating within-S transfer" heuristic is not by itself a proof. For proper coalitions of size at least three we leave (D), (E′) open and record the conditional theorem:* on exact games, ϕ is monotone under proper nonsingleton bumps provided Lemma 5.14 holds*; singleton and grand-coalition bumps remain outside this reduction. The singleton case is settled positively by Theorem 2.2, while the grand case is settled negatively by Theorem 3.2; the necessity direction (Theorem 4.1) is also unconditional.*

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